

Machine Learning and Data Mining

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Outline

Course material

Books

- ▶ Introduction to Statistical Learning with Python

https://hastie.su.domains/ISLP/ISLP_website.pdf

- ▶ Probabilistic machine learning: An introduction

<https://probml.github.io/pml-book/book1.html>

Additional references

- ▶ Probabilistic machine learning: Advanced topics

<https://probml.github.io/pml-book/book2.html>

- ▶ Machine learning in science and society (draft)

<https://github.com/olethrosdc/ml-society-science/blob/master/book.pdf>

Course github

- ▶ <https://github.com/olethrosdc/machine-learning-neuch>

Course structure

Class time

- ▶ Lectures with demos
- ▶ Lab practice
- ▶ Group work

Assessment

- ▶ Assignments (20%)
- ▶ Group project (40%)
- ▶ Exam (40%)

Communication

- ▶ ILIAS Forum for **technical** questions
- ▶ Email for **personal** questions
- ▶ Office hours Friday 13:15-14:00 (by appointment)

Course contents

Theory

- ▶ Estimation
- ▶ Learning and generalisation

Algorithms and Models

- ▶ Bayesian networks and Bayesian inference
- ▶ Neural networks and stochastic gradient descent

Social & Scientific Aspects

- ▶ Reproducibility
- ▶ Fairness
- ▶ Privacy